time for different places—there is not even a single time for any particular place. A duration can be associated only with the movement of something, with a given trajectory.

"Proper time" depends not only on where you are and your degree of proximity to masses; it depends also on the speed at which you move.

It's a strange enough fact in itself, but its consequences are extraordinary. Hold on tight, because we are about to take off.

"NOW" MEANS NOTHING

What is happening "now" in a distant place? Imagine, for example, that your sister has gone to Proxima b, the recently discovered planet that orbits a star at approximately four light-years' distance from us. What is your sister doing *now* on Proxima b?

The only correct answer is that the question makes no sense. It is like asking "What is *here*, in Beijing?" when we are in Venice. It makes no sense because if I use the word "here" in Venice, I am referring to a place in Venice, not in Beijing.

If you ask what your sister, who is in the room with you, is doing *now*, the answer is usually an easy one: you look at her and you can tell. If she's far away, you phone her and ask what she's doing. But take care: if you look at your sister, you are receiving light that travels from her to your eyes. The light takes time to reach you, let's say a few nanoseconds—a tiny fraction of a second—therefore, you are not quite seeing what she is doing *now* but what she was doing a few nanoseconds ago. If she is in New York and you phone her from Liverpool, her voice takes a few milliseconds to reach you, so the most you can claim to know is what your sister was up to a few milliseconds ago. Not a significant difference, perhaps.

If your sister is on Proxima b, however, light takes four years to reach you from there. Hence, if you look at her through a telescope, or receive a radio communication from her, you know what she was doing four years ago rather than what she is doing now. "Now" on Proxima b is definitely not what you see through the telescope, or what you can hear from her voice over the radio.

So perhaps you can say that what your sister is doing *now* is what she will be doing four years after the moment that you see her through the telescope? But no, this does not work: four years after you have seen her through the telescope, in her time, she might already have returned to Earth and could be (yes! this is really possible!) ten terrestrial years in the future. But "now" cannot be in the future . . .